cost increases, NASA changed its mind and decided to launch *Mars Observer* on a *Titan* booster. This required an additional two-year delay and an adapter to mate the spacecraft to the new launch vehicle. If NASA had not been so intent on providing business for the increasingly uneconomic shuttle, we could have launched a couple of years earlier and maybe with two spacecraft instead of one.

But whether in single launches or in pairs, the spacefaring nations have clearly decided that the time is ripe to return robot explorers to Mars. Mission designs change; new nations enter the field; old nations find they no longer have the resources. Even already funded programs cannot always be relied upon. But current plans do reveal something of the intensity of effort and the depth of dedication.

As I write this book, there are tentative plans by the United States, Russia, France, Germany, Japan, Austria, Finland, Italy, Canada, the European Space Agency, and other entities for a coordinated robotic exploration of Mars. In the seven years between 1996 and 2003, a flotilla of some twenty-five spacecraft—most of them comparatively small and cheap—are to be sent from Earth to Mars. There will be no quick flybys among them; these are all long-duration orbiter and lander missions. The United States will refly all of the scientific instruments that were lost on *Mars Observer*. The Russian spacecraft will contain particularly ambitious experiments involving some twenty nations. Communications satellites will permit experimental stations anywhere on Mars to relay their data back to Earth. Penetrators screeching down from orbit will punch into the Martian soil, transmitting data from underground. Instrumented balloons and roving laboratories will wander over the sands of Mars. Some microrobots will weigh no more than a few pounds. Landing sites are being planned and coordinated. Instruments will be cross-calibrated. Data will be freely exchanged. There is every reason to think that in the coming years Mars and its mysteries will become increasingly familiar to the inhabitants of the planet Earth.